THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 07

Full Syllabus (Chapters 1–13) | Class 9 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** tiny differences in conditions grow over time	sensitivity to initial conditions
2	**(c)** a smooth sphere layered into distinct regions	a model of layers
3	**(a)** 15 %	$\frac{6}{40} \times 100 = 15$
4	**(d)** decreases with an increase in temperature	why a warm soda goes flat
5	**(a)** an electrically neutral entity of more than one atom capable of independent existence	definition
6	**(a)** 17 u	14 + 3×1
7	**(d)** 20	neutrons = 40 − 20
8	**(b)** 4 m s⁻²	(20 − 0) ÷ 5
9	**(b)** the shortest straight-line change in position with a direction	vector quantity
10	**(b)** 100 N	F = ma
11	**(a)** act on different objects	third law
12	**(a)** a stretched bow before release	deformation stores energy
13	**(d)** 2 m	λ = v/ν
14	**(a)** animal cells	animals have no rigid wall
15	**(c)** shrink as it loses water	water leaves by osmosis
16	**(c)** xylem	tubular, thick-walled
17	**(a)** vascular tissue	dependent on moisture
18	**(b)** stamen	anther + filament
19	**(c)** ovulation	mid-cycle
20	**(a)** atmosphere	N_2 gas is 78%

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	$14 \times 60 \times 24 = 20160$ — about 20 thousand breaths (2).	1+1
22	Strong electrostatic force between oppositely charged ions (1); they form a regular crystal lattice needing much energy to break (1).	1+1
23	Velocity (rate of change of position) (2).	2
24	The exhaust gases pushed down exert an equal upward force on the rocket, larger than its weight (2).	2
25	They release energy from food during respiration and store it as ATP (2).	2
26	Xylem carries water and minerals; phloem carries food (1+1).	1+1
27	The ovary becomes the fruit (1); the ovule becomes the seed (1).	1+1
28	The cycling of matter between living and non-living parts of the Earth (2).	2

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Sublimation: solid → vapour directly on gentle heating (1); camphor deposits as solid on the cool funnel wall (deposition) (1); sand remains (1).	1+1+1
30	Their valence shells are complete (octet, or duplet for helium) (1); Ne 2, 8; Ar 2, 8, 8 (2).	1+2
31	½mv_1^2 = ½$(4m)v_2^2$ ⇒ $v_1^2 = 4v_2^2$ (2); $v_1 : v_2 = 2 : 1$ (1).	2+1
32	Air: 1 s (1); steel: 0.068 s (1); difference 0.932 s (1).	1+1+1
33	The epidermis is replaced by cork from cork cambium (1.5); cork protects and reduces water loss (1.5).	1½+1½

34	Primitive chordates with a notochord at some stage of life (2); Amphioxus (1).	2+1
35	All matter is made of atoms (1); atoms are indivisible and are neither created nor destroyed in reactions (1); atoms of an element are identical, of different elements different; they combine in simple whole-number ratios (1)	1+1+1
	OR: 12 g C → 44 g CO_2; 2.4 g C → $\frac{44}{12} \times 2.4 = 8.8$ g (2); Law of Conservation of Mass / Constant Proportions (1).	2+1
36	The atmosphere absorbs, scatters and reflects part of the radiation (3).	3
	OR: Day: land heats faster so air rises and cool sea air moves in (1.5); night: reverse (1.5).	1½+1½

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) H: 1; O: 2,6 (1). (ii) O shares one electron with each of two H atoms giving two single bonds; each H gets duplet and O gets octet (2). (iii) A compound has properties different from its elements (1). (iv) 1 : 8 (1).	1+2+1+1
	OR: (i) $6.92 - 2.48 = 4.44$ g (2). (ii) Law of Conservation of Mass (1). (iii) Final mass would appear less (1). (iv) CO_2 would escape (1).	2+1+1+1
38	(i) 2.5 m s^{-2} (1). (ii) 2000 N (1). (iii) 125 m (2). (iv) 500 m (1).	1+1+2+1
	OR: (i) Balanced: net force zero; unbalanced: non-zero (2). (ii) 100 N towards the first team (2). (iii) First team, as net force is in its direction (1).	2+2+1
39	(i) Lenses magnify an illuminated specimen (2). (ii) Resolution: clarity; contrast: difference in brightness (2). (iii) 1000X (1).	2+2+1
	OR: (i) Water: gains weight; salt: shrinks (1). (ii) Water moves from dilute to concentrated solution through the membrane (2). (iii) For a fair comparison (1). (iv) No net change (1).	1+2+1+1
40	(i) Producing new individuals of one's own kind (1). (ii) Number of parents, variation, speed, examples (3). (iii) Mitosis copies chromosomes exactly (1).	1+3+1
	OR: (i) Hydra/yeast; Rhizopus; potato/Bryophyllum (3). (ii) Single parent and identical offspring (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER